

Data Communications and Networking

Online Quiz Chapter 4

T2.2023-2024

1. Compute the signal-to-quantization noise power ratio (S/N) for 4-bit binary PCM with a sinusoidal analog signal. for $n = 8$.
 - a. 49.76
 - b. 37.76
 - c. 25.76
2. The number of bits per sample used in a standard telephony voice transmission system is 8. Calculate the number of quantized levels in PCM.
 - a. 16
 - b. 8
 - c. 256
3. Determine Nyquist interval for an analog signal represented by $s(t) = 5 \cos(100 \pi t) + 10 \cos(2000 \pi t)$.
 - a. 20ms
 - b. 1ms
 - c. 0.5ms
4. Telephone companies digitize voice by assuming a maximum frequency of 4KHz. The sampling rate is equal to.
 - a. 2000 sample/s
 - b. 4000 sample/s
 - c. 8000 sample/s
5. Asynchronous transmission means.
 - a. asynchronous at bit level
 - b. asynchronous at byte level
 - c. asynchronous at symbol level

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6. Companding means.

- a. reducing instantaneous voltage amplitude for large values
- b. Increasing instantaneous voltage amplitude for small values
- c. reducing instantaneous voltage amplitude for small values

7. For maximum audio input frequency, determine the minimum sampling.

- a. $f_s \geq 4 \text{ kHz}$
- b. $f_s \geq 8 \text{ kHz}$
- c. $f_s \leq 8 \text{ kHz}$

8. Determine the PCM transmission data rate for a single-channel PCM system employing a sample rate of 8000 samples per second and an 8-bit compressed PCM code.

- a. 1000 kbps
- b. 64 kbps
- c. 64 Mbps

9. Consider an analog signal given by $s(t) = 2 \cos(2\pi 100t) \cos(2\pi 10t)$ is sampled at the rate of 250 samples per seconds. Determine the maximum frequency component present in the signal.

- a. 220 Hz
- b. 90 Hz
- c. 110 Hz

10. for short distance usually we use.

- a. parallel transmission
- b. series transmission
- c. synchronous transmission

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